

# Publication retrain: the verdict

Fixed operating rule: 5% false alarms on 1,120 separate human controls.

## PUBLISHED HUMAN ARTICLES | FAIL

False alarms: v6 64/150 → v8 77/150 (51.3%). Goal: at most 15/150 (10%).

AI articles caught: 150/150 (100.0%). Goal: at least 95%.

Human articles with  $\geq 10\%$  of tokens wrongly highlighted: v6 38/150 → v8 52/150.

## MIXED TEXT HIGHLIGHTING | FAIL

AI tokens highlighted / human tokens falsely marked:

LLMTrace test: 34.3% / 0.7%

Synthetic v4 validation: 79.6% / 0.1%

AITDNA collaboration: 90.6% / 17.2%

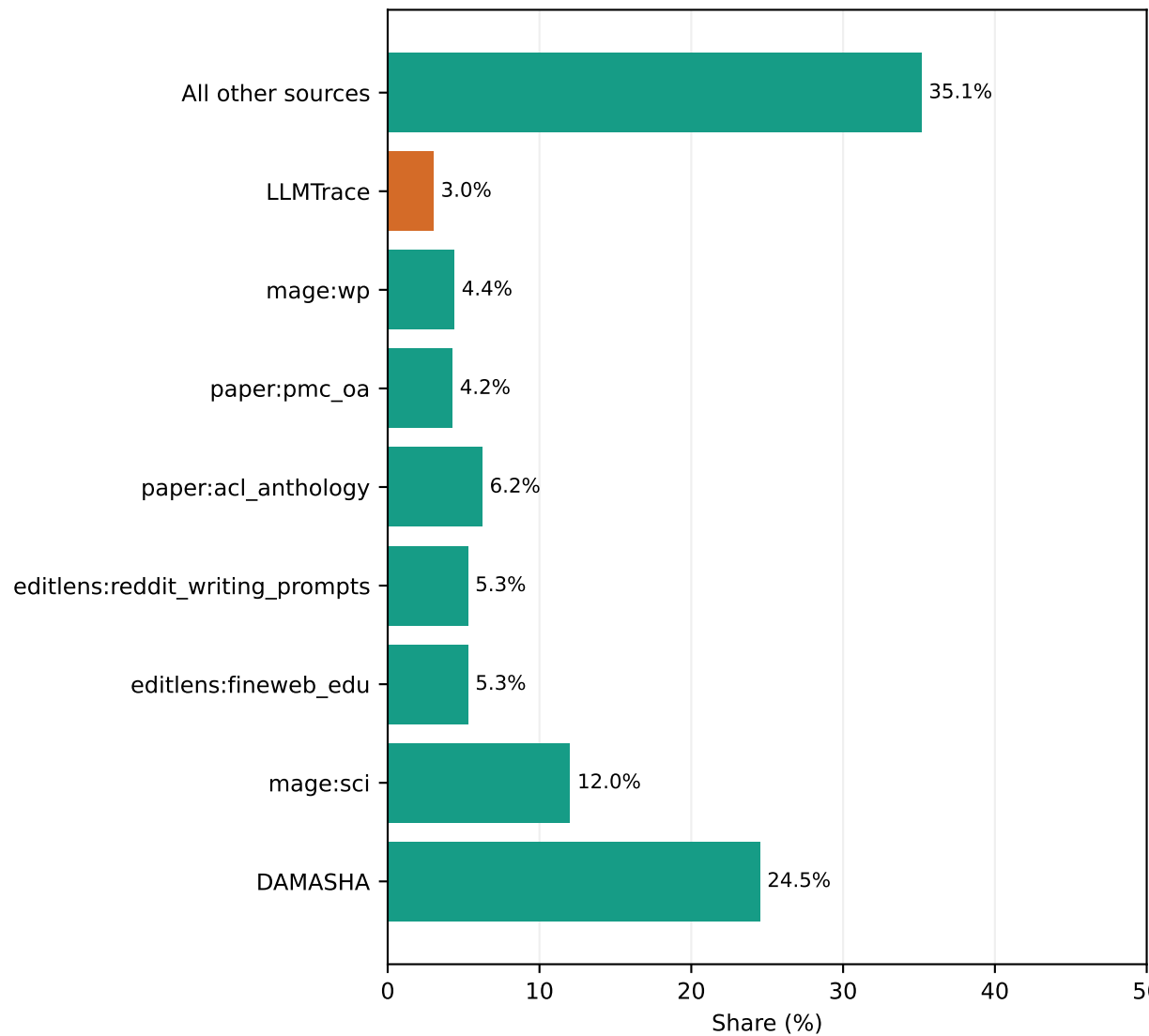
CoAuthor collaboration: 8.2% / 0.4%

Mixed-text verdict requires  $\geq 75\%$  AI-token recall on LLMTrace,  $\leq 5\%$  human-token false marks on AITDNA, and  $\geq 50\%$  AI-token recall on CoAuthor.

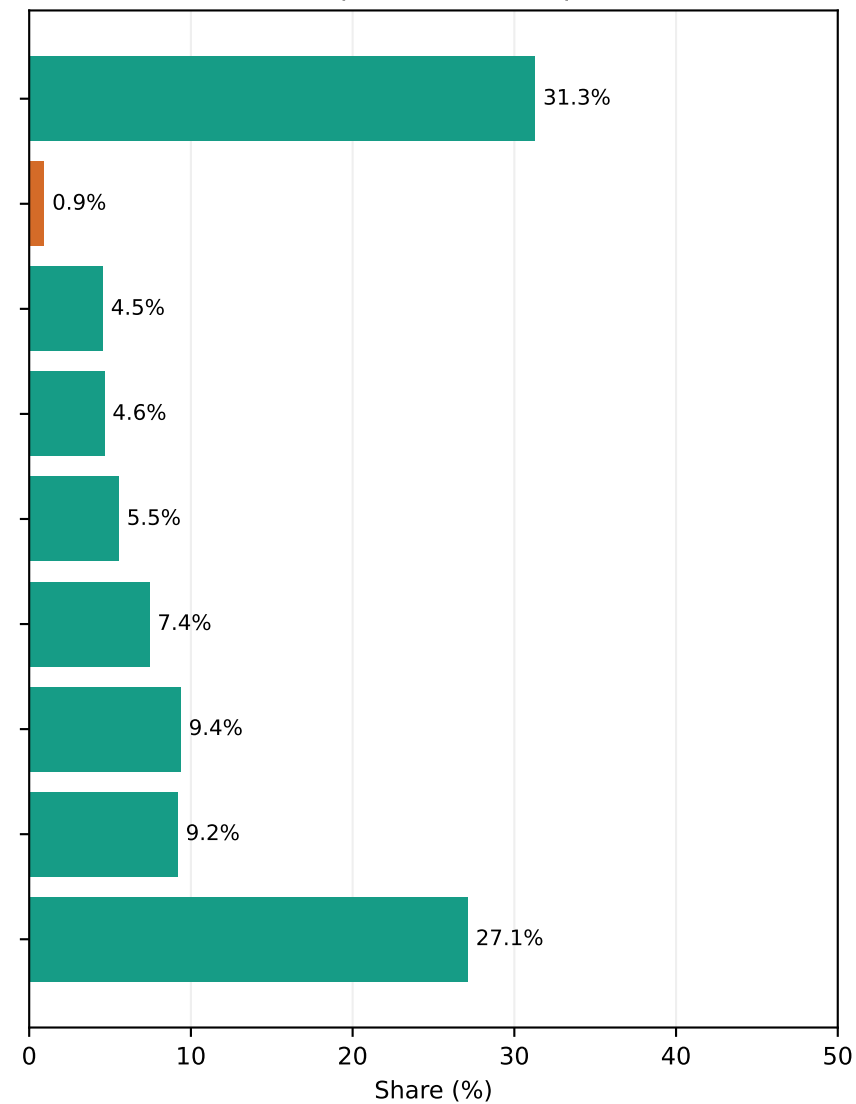
The external article set has attributed authors, but independent verification of AI-free workflows is unavailable.

# Training mix: no one source dominates

Share of 20,000 documents



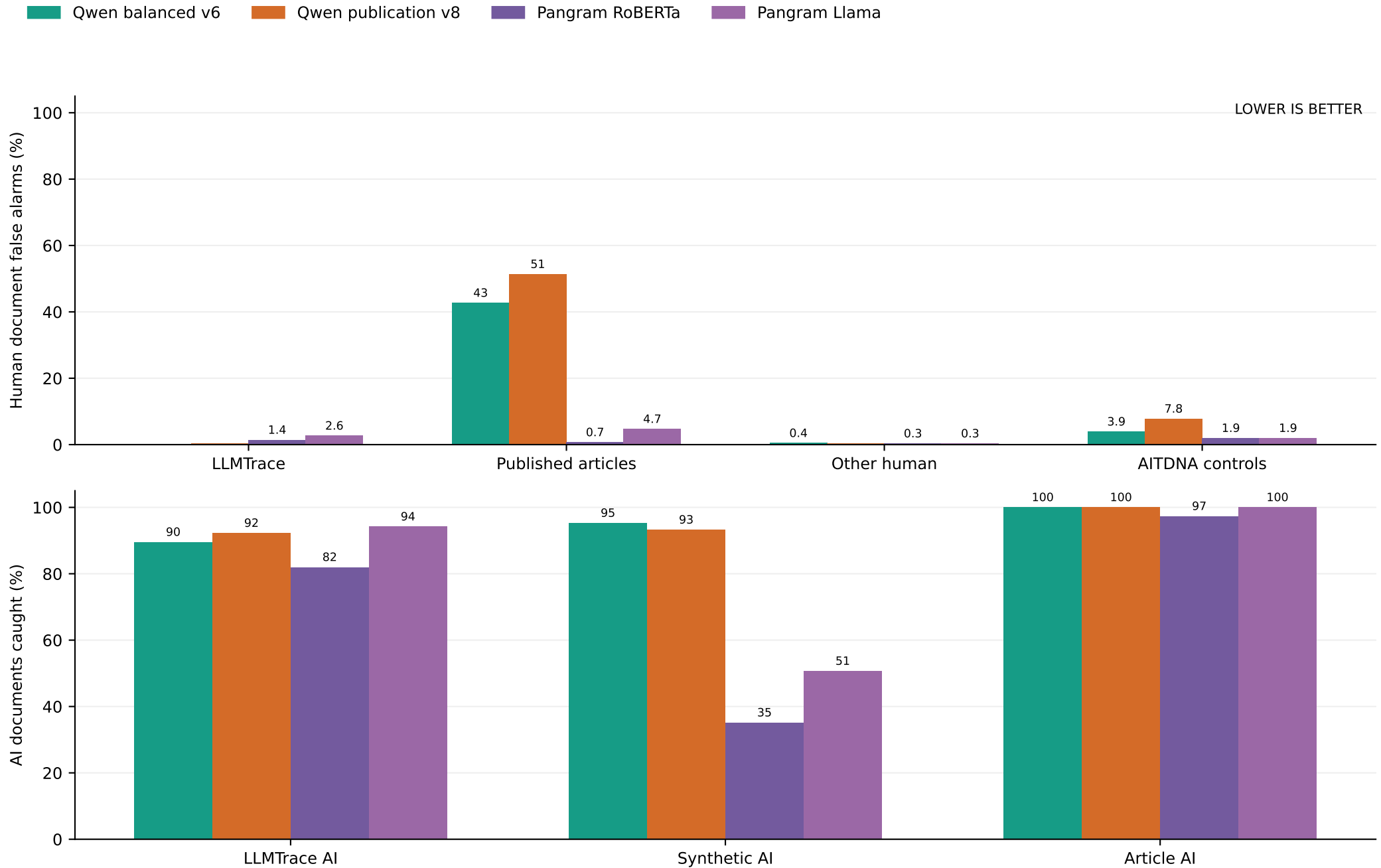
Share of supervised token positions



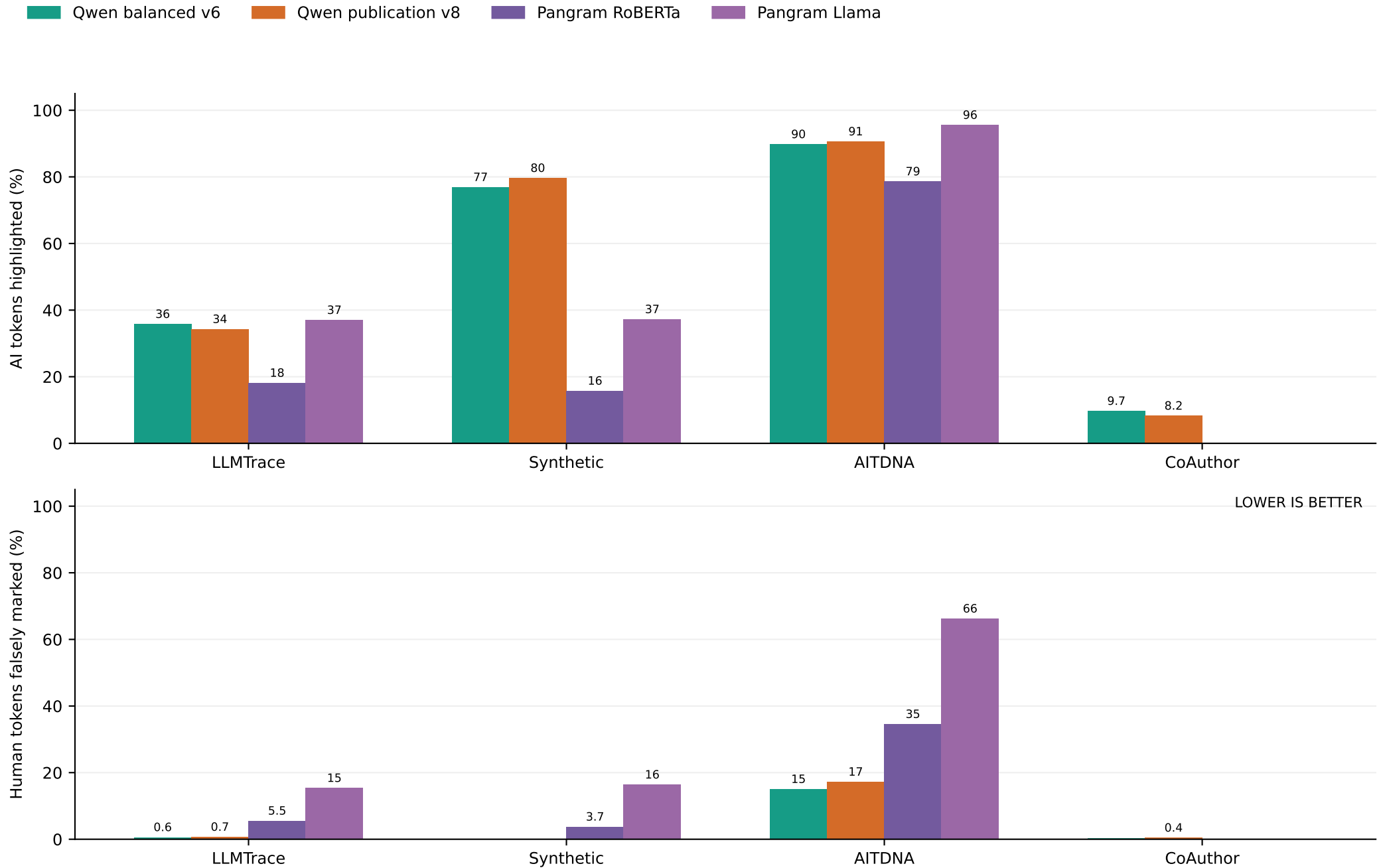
LLMTrace: 600 documents = 3.0%; 0.92% of supervised token positions. The remaining token labels are 57.1% human and 42.9% AI.

# Did publication hard negatives lower false alarms?

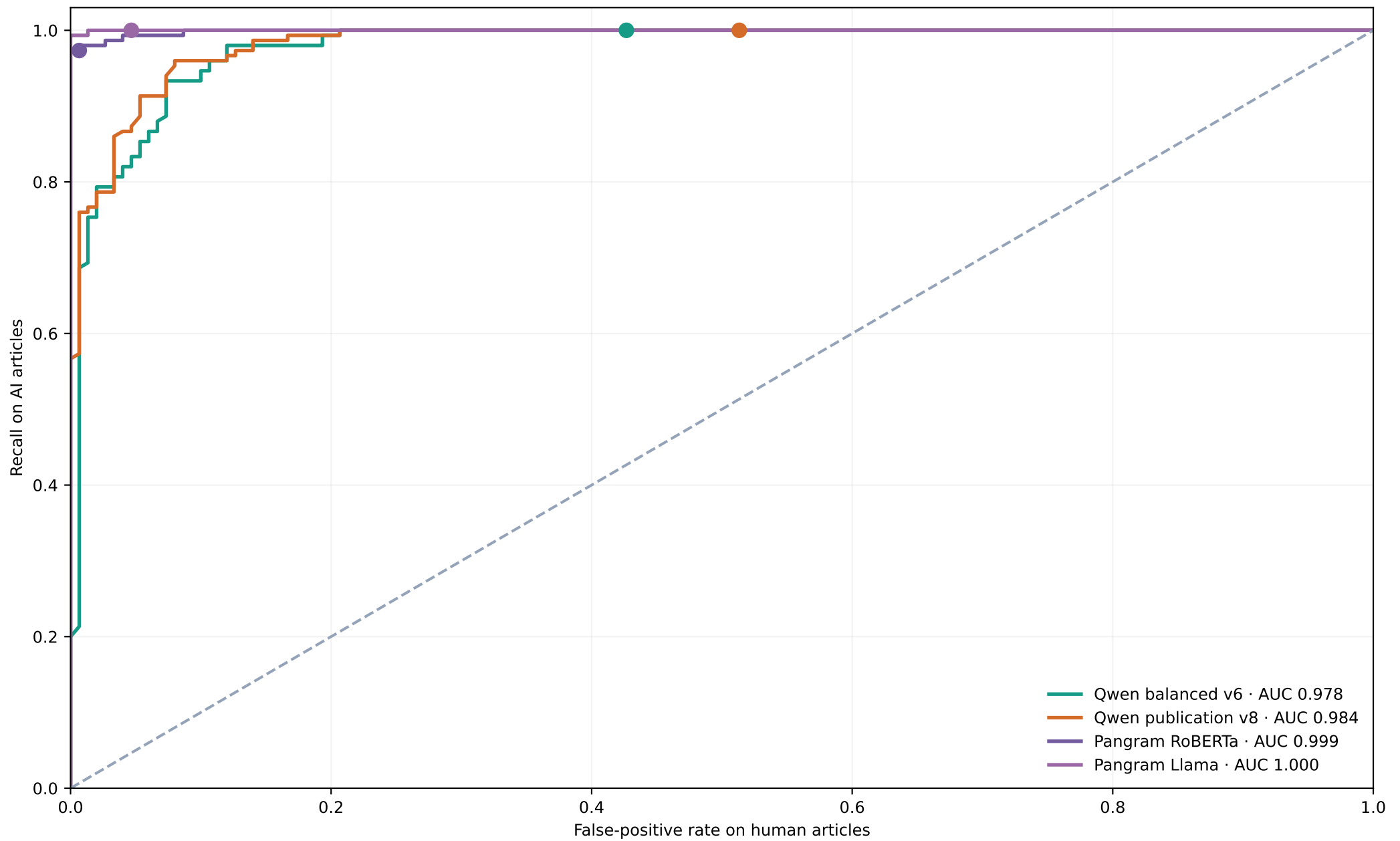
Each Qwen threshold was calibrated to 5% false alarms on the same separate 1,120 human controls.



# Mixed documents: passage detection and false marks



# Published article ROC: ranking across thresholds



# Which human publications are falsely flagged?

